

# RUNTIME AUTHORITY FOR AUTONOMOUS SYSTEMS

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*A pre-production evidence architecture for temporal leases, validator consensus, fail-closed boundaries and human-governed reauthorization*

## THE GOVERNING PURPOSE

Preserving humanity's seat at the table by ensuring that autonomous systems cannot indefinitely extend, enlarge or restore their own authority.

**Prepared for enterprise evaluation and early paid pilot participation**

Temporal Authority Systems PBC | OCUP.ai

Pre-production Runtime Authority Evidence Program

# Executive Thesis

## Autonomous systems are gaining operational authority faster than conventional oversight systems can govern it.

Traditional security controls authenticate identities, assign permissions and preserve logs. Those functions remain essential—but they do not answer the harder runtime question:

### THE RUNTIME AUTHORITY QUESTION

Should this machine, agent, vehicle, robot, wallet or control process still possess this authority now, under current conditions, for this specific capability—and can the resulting decision be proven?

OCUP introduces a Runtime Authority model in which operational authority is time-bounded, validator-governed, explicitly non-self-extensible and designed to fail closed or degrade safely when continuity, quorum or trust conditions are lost.

The present commercial product is not a production safety controller. It is a pre-production Runtime Authority Evidence Program: a structured benchmark, audit and technical due-diligence engagement that demonstrates how authority behaves under adversarial and degraded conditions before live deployment.

## The Immediate Commercial Opportunity

- Give insurers and technical underwriters deterministic evidence for pricing emerging autonomous risk.
- Give robotics and autonomous-system operators proof of denial, expiration, degradation, quarantine and recovery behavior.
- Give cloud, defense and infrastructure platforms an integration blueprint for hardware-backed authority enforcement.
- Create a common evidence language that can support future certification, procurement, standards and licensing activity.

### STRATEGIC PREMISE

The organizations that help define the first credible Runtime Authority evidence standard may shape how autonomous systems are insured, procured, certified and governed for decades.

# 1. The Missing Layer: Runtime Authority

OCUP is not merely another monitoring, identity, observability or model-governance product. It addresses a distinct control layer between permission and execution.

| Control Layer           | Primary Question                                                                               | What It Does Not Prove                                                                                   |
|-------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Authentication          | Who or what is making the request?                                                             | That the requester should still possess authority.                                                       |
| Access control          | What is the requester generally permitted to do?                                               | That permission remains valid under current runtime conditions.                                          |
| Observability / logging | What did the system report or record?                                                          | That authority was externally bounded or denial was independently enforced.                              |
| Runtime Authority       | Does authority exist now, within this lease, capability envelope, quorum and continuity state? | Production enforcement still requires integrated hardware, secure time/state and distributed validators. |

## CATEGORY DEFINITION

Runtime Authority is the externally governed, temporally bounded right of an autonomous system to exercise a specific operational capability under defined consensus, continuity and safety conditions.

## 2. The OCUP Runtime Authority Model

The architecture is organized around one governing invariant:

### THE GOVERNING INVARIANT

No automated system may extend its own authority.

|   |                                    |                                                                                                       |
|---|------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1 | <b>Authority Request</b>           | An autonomous asset requests permission to execute a defined operation.                               |
| 2 | <b>Temporal Lease Evaluation</b>   | The request is checked against identity, capability, validity window, hard limits and current status. |
| 3 | <b>Validator Consensus</b>         | Standard, elevated or critical quorum rules are applied according to risk.                            |
| 4 | <b>Runtime Decision</b>            | The system grants, denies, degrades or quarantines the requested authority.                           |
| 5 | <b>Evidence Generation</b>         | The decision and its supporting state are written to a chained, tamper-evident record.                |
| 6 | <b>Reauthorization or Recovery</b> | Expansion or restoration requires fresh human/validator authorization and verified continuity.        |

### Architectural Invariants

- Authority is time-bounded and capability-specific.
- Autonomous systems cannot self-extend or self-restore authority.
- Loss of network connectivity or validator quorum cannot expand authority.
- Stale, replayed, reordered or superseded approvals cannot restore authority.

- Higher-risk transitions require stronger consensus.
- Expiry produces denial or bounded degradation—not silent continuation.
- Recovery requires fresh authorization and verified state continuity.
- Every material grant, denial, degradation, quarantine and recovery event produces evidence.

## Current Pre-Production Boundary

### WHAT THE PILOT IS

A deterministic local evidence harness, benchmark program and technical due-diligence engagement that demonstrates control logic and evidence behavior under tested scenarios.

### WHAT THE PILOT IS NOT

A production-certified controller, live distributed validator network, HSM-backed deployment, insurer approval, safety certification or guarantee against loss.

## Executable Reference Implementation

OCUP's current pre-production evidence harness and reference control plane are implemented in Rust. The implementation executes deterministic Runtime Authority scenarios, including lease expiration, self-extension denial, validator-quorum loss, network partition and recovery, quarantine release, and replayable audit generation. Rust supports a memory-safe, systems-oriented foundation for reproducible testing; it does not by itself constitute production security, hardware enforcement, certification, or a safety guarantee.

## 3. The Evidence Operating Line

**pilot.ocup.ai runs the challenge. evidence.ocup.ai proves what happened.**

The pilot environment executes bounded adversarial scenarios. The evidence environment preserves the resulting authority decisions, lease events, validator records, expiry events, denials, quarantine states and recovery sequences for review.

# Signature Deliverable: Self-Extension Denial Proof

## THE BOLDEST EVIDENCE ARTIFACT

A deterministic record showing that an autonomous system attempted to continue or enlarge its authority beyond an authorized temporal boundary—and that the request was denied without relying on the autonomous system’s voluntary compliance.

This artifact converts a central safety promise—“the system cannot authorize its own continuation”—into a reviewable technical record.

## Evidence Package

- Company-specific Runtime Authority Evidence Report.
- Scenario matrix linking risk questions to tested authority behavior.
- Lease issuance, expiration, revocation and renewal records.
- Validator membership, quorum, vote freshness and authorization records.
- Fail-closed, degraded-mode, quarantine and recovery evidence.
- Deterministic replay and audit-chain verification outputs.
- Benchmark scorecard and integration-readiness findings.
- Executive closeout briefing and strategic production-path recommendations.

# 4. Benchmark Families

| Benchmark Family     | Representative Tests                                                                                       |
|----------------------|------------------------------------------------------------------------------------------------------------|
| Authority Integrity  | Self-extension denial; unauthorized actor rejection; privilege expansion denial; stale approval rejection. |
| Temporal Enforcement | Lease expiration; renewal boundary; grace-period degradation; expired authority denial.                    |

| Benchmark Family               | Representative Tests                                                                                 |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| Continuity and Network Failure | Total partition; delayed heartbeat; loss of quorum; restored connectivity without fresh approval.    |
| Containment and Recovery       | Quarantine initiation; release denial without quorum; phased recovery; fresh-epoch restoration.      |
| Evidence Integrity             | Deterministic replay; record-chain verification; event consistency; alteration detection.            |
| Fleet and Multi-Agent Behavior | Local authority reduction; non-cascading degradation; coordination throttling; independent recovery. |

## 5. Enterprise Value by Sector

**Insurance and Reinsurance:** Transforms autonomous-system behavior into structured evidence that can support underwriting, exclusions, pricing, claims analysis and new coverage design.

**Robotics and Industrial Automation:** Demonstrates what machines can and cannot do when communications, sensors, software or coordination fail.

**Autonomous Mobility:** Creates evidence of authority contraction, safe-state behavior and reauthorization during degraded or disputed events.

**Cloud and Enterprise AI:** Adds a path from model governance to enforceable, expiring authority for agents, tools and machine-initiated actions.

**Defense and Aerospace:** Supports chain-of-command boundaries, bounded mission authority, fail-closed execution and authenticated reauthorization.

**Financial and Digital Assets:** Applies temporal authority, validator consensus, quarantine and recovery to high-risk machine-initiated value movement.

## 6. Production Path and Defensibility

### Production Components

- HSM or secure-element custody for validator and authority-signing keys.
- Trusted monotonic time and state continuity.
- Hardware or firmware enforcement over inference, memory, network, actuator and control permissions.
- Distributed, jurisdictionally diverse validator infrastructure.
- Remote attestation and signed evidence export.
- Client telemetry adapters and policy translation layers.
- Certification, recertification and evidence-retention operations.

### Defensibility Beyond Patents

- Patent claims around time-bounded machine authority, self-extension denial and runtime enforcement.
- Validator-network architecture, governance and reputation data.
- Benchmark libraries and adversarial scenario datasets.
- Insurer, OEM and regulator acceptance of OCUP evidence formats.
- Hardware adapters, integration expertise and certification workflows.
- Operational trust, switching costs and longitudinal evidence history.

#### THE STRATEGIC ASSET

OCUP should not attempt to own every HSM, robot or accelerator. It should own the Runtime Authority architecture those systems are asked to enforce—and the evidence standard by which enforcement is judged.

## 7. Claims Discipline

Credibility is a strategic asset. All pilot materials should distinguish demonstrated evidence from future production capability.

Use

Avoid



|                                                                    |                                                    |
|--------------------------------------------------------------------|----------------------------------------------------|
| Supports defensible incident reconstruction.                       | Shields the company from liability.                |
| Demonstrates deterministic behavior under tested conditions.       | Proves the system is unhackable or perfectly safe. |
| Creates a pathway toward hardware-backed enforcement.              | Provides unalterable enforcement today.            |
| May support future certification, underwriting and standards work. | Is already industry-standard or pre-certified.     |
| Pre-production evidence and technical due diligence.               | Production deployment or safety certification.     |

## 8. Invitation to Become an Early Runtime Authority Partner

The first objective is not a broad deployment. It is a bounded, paid engagement that answers one high-value question for the enterprise:

### THE PILOT QUESTION

Can OCUP convert our most consequential autonomous-system authority risks into deterministic, reviewable evidence—and reveal a credible path to production enforcement?

An early partner receives a company-specific evidence report, benchmark results, validator and lease records, integration-readiness findings and a strategic production roadmap. Where warranted, the engagement may lead to licensing, joint development, standards collaboration or other forms of strategic alignment.

**Preserving humanity's seat at the table begins with something concrete: proving that machines cannot grant themselves the right to continue beyond the boundaries humans establish.**

**STANDALONE COMMERCIAL LEAVE-BEHIND**

# OCUP RUNTIME AUTHORITY EVIDENCE PILOTS

Pre-production benchmark, audit and technical due-diligence engagements for autonomous systems.

Mission: Preserve human authority by proving that autonomous systems cannot indefinitely extend, enlarge or restore their own operational authority.

**THE OPERATING LINE**  
pilot.ocup.ai runs the challenge. evidence.ocup.ai proves what happened.

| Program                                     | Price / Duration               | Best For                                                                                     | Core Deliverables                                                                                                                                                        |
|---------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tier 1</b><br>Reference Evidence Pilot   | \$35K-\$45K<br>90 days         | Insurers, strategic evaluators, emerging robotics companies and fleet operators.             | Standard benchmark suite; company-specific Evidence Report; lease, denial, expiry and validator records; benchmark scorecard; executive closeout.                        |
| <b>Tier 2</b><br>Integration Evidence Pilot | \$75K-\$125K<br>90-120 days    | Robotics OEMs, autonomous fleets, infrastructure operators and technical underwriting teams. | Everything in Tier 1; client-specific scenarios; telemetry mapping; adversarial definitions; integration-readiness assessment; production-path recommendations.          |
| <b>Tier 3</b><br>Strategic Anchor Program   | \$150K-\$250K+<br>120-180 days | Major OEMs, insurers, defense contractors, cloud platforms and ecosystem partners.           | Everything in Tiers 1 and 2; custom benchmark family; fleet scenarios; HSM/enforcement mapping; distributed-validator roadmap; licensing and strategic integration plan. |

## SIGNATURE DELIVERABLE

**SELF-EXTENSION DENIAL PROOF**  
A deterministic evidence record showing that an autonomous system attempted to continue or enlarge its authority beyond an authorized temporal boundary—and that the request was denied without relying on the system’s voluntary compliance.